A Covariant Screened Theory for a Dark-Energy-Tied Galaxy Law: Both Covariant Forms Are Ill-Posed Somewhere, and the Khronon Form Fails Growth, Hosted Systems and Cluster Cores

Working note, 2026-10-23, v1.1 (corrects the cause of v3's growth failure and adds the v3 recompute lanes; see "Changes in v1.1" at the end). AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ FITTED; the theory is not closed.

Abstract

The owner's programme studies a gravity framework whose galactic law is $g_N = g^2/(g+a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ FITTED; its extra gravitating mass is a collisionless "cold energy" medium, not a particle. The companion note PAPER58 found that only a screened version, with a stiffness length $L_\star$, passes both Cassini and the galactic law. This note writes that screened theory covariantly in two forms and tests both: v3, the Blanchet–Skordis Khronon with an independent spatial covector and one stiffness term γ_L, and v4, AeST with the same term. Every lane froze its criteria before computing and ran a mutation control; every lane added in v1.1 was rerun by the coordinator. Both a_0 footings (9.36×10^{-11} and $1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled.

In plain terms. The covariant candidate v3 is ill-posed wherever the khronon density is negative: at every two-body saddle and, with an external field, in Mpc-wide lobes around every system. The γ_L screening term is what makes that instability unbounded (H4). v4 is ill-posed wherever the aether twists (I1). v3 also fails several structural tests: linear growth at the dust mass that galaxies allow, a radiation era in its literal form, hosted dwarf satellites, Coma's ultra-diffuse galaxies, the Milky Way's outer slope, and cluster cores for every allowed dust mass. It passes Cassini and the planets (screened), the strong-field tests (scoped), the MeerKAT a_0, isolated groups, and, natively and with no knob, tidal dwarfs and DF2/DF4. What the data require of any successor is a cold component that binds in hosted systems and in clusters, and grows like dust.

1. **Erratum (M1).** v3's linear growth failure is caused by Blanchet and Skordis's cap on the dust mass μ_K, not by $J'(0) = -1$ as v1.0 stated. At the cap J_x holds the dust better than $J'(0) = 0$ (σ_8 ratio 0.391 against 0.121); reaching a ratio of 0.95 needs $\mu_K \geq 411\,h$/Mpc, 27.8 times the cap.
2. **Cosmology (Q6, Q7, C7).** Literal v3 has no radiation era: BBN needs $\mu_K \geq 1.84 \times 10^8\,h$/Mpc. BS25's DBI K fixes BBN only: growth stays at $R_B = 0.26$–0.32 of the dust limit. v4 runs away on FRW for the dust shape tried.
3. **Hosted systems (S1, U1, T1).** v3's own dust obeys two theorems (U1): held dust never exceeds the AQUAL phantom (T1), and it is weightless in a static host (T3), so hosted systems are dust-free. Satellites then sit +2.00 / +1.99 dex and Coma's UDGs +1.41 dex above v3: STRUCTURAL. Tidal dwarfs and DF2/DF4 become a NATIVE PASS. One external-field strength does not fit the data (T1).
4. **Galaxies and the Milky Way (P4, S3).** SPARC is CONDITIONAL. The Milky Way's outer slope is SHAPE EXCLUDED ($Z = -4.37$ / -4.62). κ is CONSISTENT, NOT DISCRIMINATING. A v3 Gaia DR4 Milky-Way file and a v3 ALMA reading are sealed.
5. **Clusters fail, in both forms (F1, H2, S4).** Rebuilt from the raw eRASS1 and KiDS data, the cluster cores sit 1.68–1.85 (canonical) and 1.54–1.70 (alternative) times above the law on their baryons, and no μ_K that galaxies allow fixes them.
6. **What passes.** Cassini (all 24 cells, by 4.8×10^7 to 5.1×10^9), the planets, the outer globulars (which set $L_\star \geq 21.68$ / 22.85 pc), and the strong field: PASS on all five tiles, scoped to static non-rotating black holes, with strong coupling an estimate (P3).
7. **Pending.** The Khronon-native nonlinear production run (L1) is in progress; no result is given. Gaia DR4 is the sealed decisive test of screening: screened $\gamma_v - 1 \approx 8.1 \times 10^{-6}$ against $\gamma_v \approx 1.17$ for the unscreened reading.

1 Setting and rules

- $\kappa = \frac{1}{2}$ FITTED. The thermal value $\kappa = Z/4\pi$ is Milgrom's relation $2\pi a_0 \approx c^2(\Lambda/3)^{1/2}$ (arXiv:1605.07458, eq. 5), a postulate here (PAPER58). Footings: canonical $a_0 = 9.36 \times 10^{-11}$, alternative $1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled.
- Status words follow the record: DERIVED (from listed premises), COMPUTED, POSTULATED, FITTED, OPEN, PENDING. The source document is `covariant_push_2026_10_22/THEORY_V3_DRAFT.md`: commit 5d7308f3a2 for v1.0 and commit be52681b9b for v1.1. Every number added in v1.1 is taken from that draft's rows and checked against the lane's README. Lanes C1–C8 are in `covariant_push_2026_10_22/`; D10 and E6 in `doors_push_2026_10_22/`; F1 onward in `rebuild_2026_10_22/`.
- **Two forms.** v3 is the Khronon form of the record's theory draft; v4 is the AeST form found in the rebuild lanes (F1–H3), added to the draft as its §1b (commit 8bcafea2fa). The rebuild lanes were rerun by the coordinator;

Table 1: Lanes of v1.0: the covariant push, two late doors lanes and the first rebuild lanes. Commits: criteria / result. [a]Frozen rule REAL FAILURE; not robust, so UNDECIDABLE under a post-result amendment. [b]Superseded in v1.1 by M1 (cause) and Q6, Q7 (size).

lane	question	verdict	checks	commits
C1	AeST embedding	EMBEDS WITH EXTRA TERM	8/8	d9a9a17f4e / 6502ef90a9
C2	AeST cosmology, native H_0	COMPUTED + DATA-GATED	9/9	ac4b30ff62 / 2dd388643a
C3	AeST scales vs $L_\star$	SCOPED NO-GO	9/11	c16ef33ae5 / fd7196584c
C4	Khronon parent	EMBEDS WITH EXTRA TERM	8/8	b0ea16ecf1 / 5f4f845554
C5	health	CONDITIONALLY HEALTHY	7/10	f002ce909a / f3e7d4d661
C6	Cassini, literal form	UNDECIDED (frozen)	7/8	1b2ced55a4 / ae428dc252
C7	native CMB code	CRISP FINDING	7/7	milestones to 64b0fdc9b4
C8	instantaneous sector	WELL-POSED + UNCONSTRAINED	8/8	f2399aaf5e / aa113a5e96
D10	native $L_\star$ floor	CRISP FINDING	10/10	e3333cb274, A1 ecec1ce9e7 / 3ac0573063
E6	native catchment	CRISP FINDING	5/5	c42925d9dc, A1 5c0669a33d / 0bea588820
F1	cluster cores, rebuilt	REAL FAILURE	6/6	9fd5aee16e / f1803b0256
F2	outer lensing, rebuilt	UNDECIDABLE[a]	9/9	e9816b47d2 / 7c8c3d5302
F3	growth vs v3	v3 growth FAILS[b]	9/9	0bef684eea / 059766caa5
G1	old engine's excess	ARTEFACT (old engine)	7/7	9921a5172a / a048e6d468
G2	pre-infall retention	FAILS (native)	4/6	92716d0148, A1 b3fbb9ed9d / 3cdcf3b4c5
H1	AeST free function	NONE IN THE FAMILY	6/6	a4d76f492e / 1654b84f78
H2	AeST cluster mass	CANNOT	6/6	364a56c250, A1 5160a89939 / 385f61a14c
H3	the health criterion	CLAIM FAILS (no ghost)	4/5	04355110c4 / 6b905cb6d8

F1 and H2 were reproduced on an independent KiDS extraction. The lanes added in v1.1 recompute v3's tiles one at a time and test its cosmology and health.

- Table 1 lists the lanes of v1.0 that are not in PAPER58; Table 2 the lanes added in v1.1, with the draft commit that added each row.

2 The v3 action (C1, C3, C4, C5)

The Blanchet–Skordis Khronon action is $S = (c^3/16\pi G)\int \sqrt{-g}\,[R - 2J(Y) + 2K(Q)]$, with a preferred foliation τ, khronon acceleration A_μ and $Y = A\cdot A/c^4$ (Blanchet & Skordis 2024, 2025; the equations were read via the arXiv HTML and re-derived symbolically in C4, not verified on disk). The screened theory is:

- **Literal form.** $\beta = 0$ (no extra curvature term), $\eta = 0$ (no kinetic term for $\mathbf{P}$), Blanchet–Skordis $K(Q)$ dust.
- **The free function.** $J = J_x = \Lambda - 2a_0^2[y - \ln(1+y)]$, $y = \sqrt{Y}/a_0$, gives $1 + J' = y/(1+y)$: AQUAL with $\mu = x/(1+x)$ exactly (C4, sympy). But $J_x/\sqrt{Y} \to -2a_0$ at large Y, so it lies outside Blanchet and Skordis's admissible class, which demands $J \to$ const. POSTULATED.
- **The addition.** $\mathbf{P}$ promoted to an independent spatial covector ($n\cdot P = 0$) with one stiffness term $-(\gamma_L/2)(D_iP^i)^2$, $\gamma_L = KL_\star^2/2$: one new field and one new constant. The pure Khronon cannot screen the Sun: with J_x the planets see the full a_0, 1.08×10^4–8.07×10^4 times their bounds; with a turn-off J the AQUAL Q_2 still fails by at least 2.44× (canonical) and 2.85× (alternative) (C4, b0ea16ecf1 / 5f4f845554).
- **Static limit.** $\phi = \psi$ (lensing sees the dynamical potential) and $\nabla\cdot[(1+J')\nabla\psi] = 4\pi G\rho$ for any mass distribution: the field equation is AQUAL, not QUMOND, matching PAPER58's E2. The curl that AQUAL needs lives in the unconstrained auxiliary P^* ($|\nabla\times P^*|/|\nabla P^*| = 0.126$ on a two-mass test), not in the foliation.
- **Relation to AeST (C1, C3).** AeST (Skordis & Złośnik 2021) reproduces AQUAL with $\mu = x/(1+x)$, the clock term and the curl half of the stiffness exactly, but the longitudinal stiffness $\gamma_L(D\cdot P)^2$ is identically zero in AeST and in any first-derivative extension; it needs second derivatives of AeST's scalar (C1, d9a9a17f4e / 6502ef90a9). No term in AeST's class screens small systems (C3, SCOPED NO-GO, c16ef33ae5 / fd7196584c): AeST's mass term acts at large r and in the opposite direction. So $L_\star$ is a new constant of the covariant theory.
- **Relation to DSF-21's A33.** A33 is this theory with $\eta > 0$. With the Khronon's curvature term ($\beta = 0$),

Table 2: Lanes added in v1.1 (all in `rebuild_2026_10_22/`, all reproduced by the coordinator). Verdicts as written in the theory draft's rows; the commit is the draft commit that added the row.

lane	question	verdict	draft row
M1	the growth trilemma	ERRATUM: the cause is the μ_K cap	0eeb77c747
Q6	v3's own CMB and $P(k)$	no radiation era (literal K)	c7cc6f9e61
Q7	BS25's DBI K	fixes BBN only	428e689b49
C7	v4 on FRW (milestone 5)	v4 RUNS AWAY (dust shape tried)	0d39e739a6
H4	v3 where $\rho_\tau < 0$	ILL-POSED	9fe0dcb5b5
I1	v4's causal structure	ILL-POSED where the aether twists	1b5444fb47
P3	strong field	PASS on all five, two scoped	c9f7f1cc09
P4	galaxy tiles	SPARC RAR CONDITIONAL; MW vertical force, TDGs, DF2/DF4 FAIL (static)	433e29eb85
S1	hosted small systems, EFE solved	both FAIL	244f4468f6
U1	what v3's own dust can supply	T1, T3; STRUCTURAL / NATIVE PASS	428e689b49
T1	external-field strength	FOOTING SPLIT, FRAGILE	dc6d6a3e1e
S2	inner and red lensing	both CONDITIONAL	bc441b4656
P2	spectroscopic outer lensing	UNDECIDABLE (N too small)	244f4468f6
S3	Milky Way, κ, seals	SHAPE EXCLUDED; CONSISTENT, NOT DISCRIMINATING	52286e05ee
S4	μ_K on clusters	SCOPED NO-GO	45bd132464
P1	dwarf supply from chemistry	SUPPLY CONFIRMED for the median, not robust at 2σ	7602bc3238
J1	potential-depth selector	FAILS	a410688c53
K1	braneworld Weyl term	FAILS	52360cde46
M2	screened tidal-charge term	FAILS	0944553f40
L1	Khronon-native PM engine	production in progress (no result)	lane README

$\eta > 0$ gives a gradient instability wherever the field is nonzero; in the Solar System at $k = 1/\mathrm{AU}$ the rate is 83.3 /yr (canonical) and 95.09 /yr (alternative). So η is dropped (C5).

- **A closure identity (C4).** If $J(\infty) = 0$ (Blanchet–Skordis 2025), the action gives $\Lambda c^4/a_0^2 = 2F(y_t)$, twice the record's closure, so the turn-off scale would be 33.94 (canonical), 23.70 (alternative) or 33.98 ($\kappa = \frac{1}{2}$). This applies only to the turn-off branch, which fails Cassini anyway, and it does not select κ.

3 Health (C5), as corrected by P3 and H4

Criteria f002ce909a, result f3e7d4d661. Verdict CONDITIONALLY HEALTHY, and only in the literal form; v1.1 narrows it (last item).

- **The literal form at leading order.** No ghost and no gradient instability on every background tested where the frozen-coefficient (WKB) expansion is valid; only the two tensor modes propagate, at c; **P** and the khronon scalar obey elliptic constraints; the $\omega = 0$ mode has positive energy at every k; the FRW dust has $c_s^2 = (\bar{Q}-1)/\bar{Q}$. Conditions: $\gamma_L > 0$; W convex (the record's W meets this at every field strength); a positive dust charge; $\beta = 0$ and $\eta = 0$ exactly.
- **What fails.** $\eta > 0$ with $\beta = 0$: ill-posed (above; the stability boundary is the record's $\beta > 2\eta a_0^2/K$, confirmed at $\beta = 1.48707 \times 10^{-19}$). $\beta > 0$ with the $K(Q)$ dust: an infrared gradient band. AeST plus the stiffness term was ranked second: its $\omega = 0$ mode has negative energy at every k in vacuum, and the stiffness term gives an unbounded speed. (H3 later showed the first defect is the energy criterion flagging ordinary dust; Section 7.)
- **Checks.** 7/10, exit 1. Three frozen checks fail and are kept: H5 (a weak-field factor fails at neutron-star compactness), H6 (all failing points lie where the background is not local; this reading is post hoc) and H9 (the $\beta > 0$ dust band, physics). Strong coupling is far below every sub-Hubble scale.
- **Corrected in v1.1.** "Only tensor modes propagate" holds only at leading WKB order: at first order in background gradients the khronon/dust pair propagates wherever $D \cdot P > 0$ (P3), and it is gradient-unstable, without bound, wherever the khronon density is negative (H4, Section 5).

4 Well-posedness and causality (C8)

Criteria f2399aaf5e, result aa113a5e96. WELL-POSED + UNCONSTRAINED, scoped to the weak-field leaf problem; 8/8 scored checks.

- **Well posed.** On each leaf the **P** and lapse equations are the Euler–Lagrange equations of one strictly convex functional, so the solution exists and is unique on a bounded domain for every source below the cap ($\gamma_L > 0$). Because the action has no $\dot{N}$ or $\dot{P}$, the time-dependent leaf problem is exactly the static one. OPEN in v1.0: the infinite-volume theorem, the coupled system with GR's tensor sector and the dust, and strong fields. **v1.1:** H4 answers the coupled system: it is ill-posed where the khronon density is negative. C8's WELL-POSED verdict stands for the leaf (**P**, lapse) problem only.
- **Criterion A is violated; criterion B is met.** The response is a real, gauge-invariant infinite-speed influence on the khronon's leaves. It never goes backward in τ, so the single-global-time criterion holds. Its size: moving 1000 kg by 1 m changes the force 1 km away by $1.2 \times 10^{-46}\,\mathrm{m\,s^{-2}}$ at $L_\star = 24\,\mathrm{pc}$.
- **No existing data constrain it.** Binary pulsars: no radiative channel, and the conservative correction is at most 3.2×10^{-14} of the double pulsar's precision. Preferred frame: an α_2-equivalent of 1.7×10^{-7} of the bound (this also corrects the record's A22 D5, which missed the leaf tilt). GW170817: $\Delta v/c \leq 6.2 \times 10^{-32}$. P3's propagating pair couples to the metric at most at 2.4×10^{-9}, so the binary-pulsar result survives.

5 v3 is ill-posed where the khronon density is negative (H4); v4 where the aether twists (I1)

- **The dispersion (H4, sympy).** In the weak field, to first order in background gradients and with self-gravity, the khronon/dust pair obeys $\omega^2 = \rho_\tau[k^2/\chi(k) - 4\pi G]$; at $\mu_K = 0$, $\omega^2 = 4\pi G\rho_\tau(M_\theta - 1 + L_\star^2 k^2/2)$. Here ρ_τ, the khronon fluid's density, is the phantom $D \cdot P$ plus the dust.
- **ILL-POSED.** Where $\rho_\tau < 0$ the growth rate $\Gamma = L_\star k\sqrt{2\pi G|\rho_\tau|}$ rises linearly with k without cap: a gradient (Hadamard) instability. It is the γ_L screening term that makes it unbounded; without it the same regions grow by only 0.006–0.20 e-folds over their lifetime.
- **Where.** $\rho_\tau < 0$ at every two-body saddle (Milky Way–M31, a region about 290 kpc across; LMC–Milky Way, 12 kpc; galaxy pairs; cluster pairs) and, with an external field, in Mpc-wide lobes around every system, single galaxies and discs included. Wavelengths below about 4–330 pc grow by more than one e-fold over the region's life. The Solar System, the Sun–Galaxy null and the Earth–Moon saddle are HEALTHY: screening keeps $D \cdot P \geq +0.045\,M_\odot\,\mathrm{pc}^{-3}$.
- **No fix inside v3.** No dust offset or μ_K removes the regions. A cutoff would have to sit near $L_\star$, but γ_L must act at AU scales to screen the Solar System. H4 corrects C5's and C8's statements above, P3's "untested" and the draft's own well-posedness row.
- **v4 (I1).** The γ_L term gives v4 a real dispersive scalar ($\omega \propto k^2$). On aether-orthogonal slices (Minkowski, FRW, exact spherical symmetry) the Cauchy problem is well posed; on any slice tilted relative to the aether two roots go complex above a critical wavenumber, so it is Hadamard ill-posed. v4 needs a twisting aether to give the law off symmetry (C1), so every non-symmetric bound system, the Sun in the Galaxy's field and discs included, has no good slice, and relayed signals can close loops in time: criterion B is violated in principle (illustrative loop gains 4 ms–3.9 s around the Galaxy). No completion was found: a $(\Box\phi)^2$ partner keeps the statics but adds an Ostrogradsky ghost, and a hypersurface-orthogonal aether is v3.
- **So the γ_L screening operator makes both covariant forms ill-posed somewhere** (H4 with I1).

6 Linear growth and cosmology of v3 (F3, M1, Q6, Q7, G1)

F3: criteria 0bef684eea, result 059766caa5; 9/9. M1, Q6 and Q7 are new in v1.1.

- **Erratum (M1): the cause is the μ_K cap, not $J'(0) = -1$.** v1.0 wrote that with J_x, $J'(0) = -1$ exactly, so the Newtonian term vanishes on FRW. That cause is wrong. The exact dust kernel from Blanchet and Skordis's post-Newtonian system is $s(k; f_0) = (\mu^2 - f_0 k^2)/((1-f_0)k^2 + \mu^2)$, $f_0 = 1 + J'(0)$: J_x ($f_0 = 0$) gives $\mu^2/(k^2+\mu^2)$ and $J'(0) = 0$ gives $1 - k^2/\mu^2$. Both tend to 1 as $k \to 0$, so the Newtonian term survives on large scales in v3 too, and J_x holds the dust better than $J'(0) = 0$ (σ_8 ratio at the cap 0.391 against 0.121). Growth fails because the dust's late Jeans scale is μ_K for every K analytic at $Q = 1$, and Blanchet and Skordis's galaxy bound caps

μ_K. Reaching $\sigma_8 \approx 0.95$ of the dust limit needs $\mu_K \geq 411\,h/\mathrm{Mpc}$, 27.8× the cap; no analytic K shape escapes.

- **The gap (F3; the numbers stand).** To keep σ_8 within 5% of the dust limit μ_K must be at least 361–2542 h/Mpc, while Blanchet and Skordis's bound caps it at 14.8 h/Mpc: a gap of 24–172×. Inside the allowed range linear σ_8 is only 0.04–0.39 of the dust limit; above it the dust forms galaxy halos and double-counts against the law.
- **Literal v3 has no radiation era (Q6).** Blanchet and Skordis's quadratic $K = \mu_K^2(Q-1)^2$ carries a stiff a^{-6} partner whose amplitude is fixed by today's dust density; it beats radiation before $z = 1854$ ($\mu_K = 14.8\,h/\mathrm{Mpc}$) up to 3.2×10^5 (2542 h/Mpc). BBN needs $\mu_K \geq 1.84 \times 10^8\,h/\mathrm{Mpc}$, 1.2×10^7 times the cap. On a dust background with the literal perturbations (not the literal theory), full Boltzmann growth at the cap gives $\sigma_8 = 0.061$ ($R_B = \sigma_8/\sigma_{8,\mathrm{dust}} = 0.073$, against F3's ODE 0.391), and 5% growth needs $\mu_K > 2542\,h/\mathrm{Mpc}$. Fitted to plik-lite, $\Delta\chi^2$ against ΛCDM runs from +13891 (not converged) down to +18.7 over that range of μ_K. These numbers test v3's dust sector, not the galactic law (a_0 is absent at linear order).
- **BS25's DBI K fixes BBN only (Q7).** The DBI form makes the early khronon dust-like (ΔN_{eff}-equivalent 4×10^{-6}–7×10^{-6} at 1 MeV, against 7.7×10^{13} for the quadratic K), but linear growth still fails: $R_B = 0.26$–0.32 at Blanchet and Skordis's own parameter sets and the cap, and plik-lite $\Delta\chi^2$ is +350 to +748. Blanchet and Skordis ran no Boltzmann code; their CMB claim compares only the background $w(a)$. A much larger DBI constant restores σ_8, but it is a new constant set by the failure, leaves $P(k)$ at 1 h/Mpc at 0.33 of ΛCDM, and would make K complex inside cluster potentials (order of magnitude); it is reported, not adopted.
- **The record's growth failures belong to the old settling rules,** not to v3. The KiDS $\Delta\chi^2 = +68.7$ (CFG596) is reproduced exactly (+68.73) and comes from the settling rules and the QUMOND phantom, which v3 does not contain. The old engine's 15–19% power excess is not even converged: it grows with the rule's grid, 0.040 / 0.077 / 0.163 at 128^3 / 256^3 / 512^3, and a smooth cap makes it worse (G1).
- **In the $\mu_K \to \infty$ limit** (the dust limit), linear $S_8 = 0.845$ / 0.814 against raw KiDS on $k \leq 0.3\,h/\mathrm{Mpc}$ (0.754 ± 0.063 / 0.761 ± 0.060): CONSISTENT. But that limit is outside the allowed range.
- **v3's one open door is nonlinear**: whether the law's own nonlinear term restores growth (L1, Section 11). M1 found no linear way through.

7 v4: AeST with the stiffness term and a capped free function (C1, C5, C7, H1, H3, I1)

- **The action.** Skordis–Złośnik AeST with C1's capped free function (its static limit is the canonical law exactly) plus the same single stiffness term γ_L, which AeST lacks (C1). POSTULATED.
- **What it passes analytically.** The Newtonian term survives in the infrared for any free function ($\Gamma^2(k \to 0) = 4\pi\tilde{G}\bar{\rho}$); v1.0 contrasted this with v3, but M1 shows v3's survives as $k \to 0$ too. The galactic law exactly; Cassini (screened, as v3); BBN ($\lambda_s = \infty$). No member of Skordis–Złośnik's own family passes all of these (H1, a4d76f492e / 1654b84f78): BBN needs $\lambda_s \geq 24.7$, where their interpolants lose to the canonical law on SPARC at $z = -2.37$ to -2.74 (canonical) and -2.78 to -3.64 (alternative).
- **The C5 health criterion was mis-specified (H3, 04355110c4 / 6b905cb6d8).** The same energy test flags ordinary self-gravitating Newtonian dust and GR plus k-essence dust. The AeST mode is non-propagating dust, ghost-free at linear order on the vacuum, with growth rate bounded by $\sqrt{4\pi\tilde{G}\bar{\rho}}$ (maximum 0.99991 of it) and a band that ends at $k_c \approx 243\,\mu$. Between μ and k_c the binding is AeST's contact-term static law, up to 2.3×10^4 times Newton near $k \approx 185\,\mu$. So v4's health held at linear order only.
- **v4 runs away on FRW (C7, milestone 5, 64b0fdc9b4).** For the dust shape tried (Cosh, $Q_0 = 0.1\,\mathrm{Mpc}^{-1}$), every v4 configuration has $P/P_{\Lambda\mathrm{CDM}}$ of 10^2–10^{41} by $k \leq 10\,h/\mathrm{Mpc}$; the result is gauge- and tolerance-stable, so not numerical. The likely mechanism is a tachyonic vector pair before $z \approx 78$. Untested escape: a different early K, or $Q_0 \gtrsim 30\,\mathrm{Mpc}^{-1}$.
- **v4 is ill-posed where the aether twists** (I1, Section 5).
- **No screening without γ_L (M1).** Any healthy AeST or A33 law that matches SPARC leaves the planets an anomaly of at least $0.96\,a_0$ (theorem T2); galileons and potential, density or Weyl keying all fail. The trilemma stands: Solar-System screening needs γ_L; γ_L is well posed only with an untwisted aether; the untwisted (Khronon) form fails linear growth at the allowed μ_K. v1.0 called v4 the better-surviving candidate; with I1 and C7 neither form is a complete, healthy theory.

8 Parameters

- a_0 **(κ): FITTED.** $\kappa_\Lambda = 0.417 \pm 0.095$ (CFG493); 0.414 ± 0.094 on a ΛCDM-free background (PAPER58, E4). Under v3's AQUAL discs (S3): $\kappa_\Lambda = 0.499 \pm 0.116$ and $\kappa_{\rm crit} = 0.413 \pm 0.096$ ($\frac{1}{2}$ at 0.01σ and 0.82σ): CONSISTENT, NOT DISCRIMINATING. Part of the shift is a record error: CFG493's v2_disc has ring-singularity noise.
- $L_\star$**: FITTED, a new constant, lower bound only.** With the theory's own field equation in the Galactic field (D10, result 3ac0573063), the floor is $L_{\star,95} = 21.68$ pc (canonical) / 22.85 pc (alternative) with the stellar-population M/L; over the M/L priors and the Milky-Way bracket it spans 2.93–22.67 / 4.40–23.80 pc and straddles 20 pc. The record's upper edge is about 90 pc (PAPER58, D8).
- **The dust abundance: FITTED.** It is an initial condition, the khronon's conserved shift charge (C4) or AeST's (C2); nothing in the action fixes it.
- $G_{\rm cosmo}/G_N$**: bounded.** BBN (the record's A78): $G_{\rm cosmo}/G_N > 0.9611$. A Planck-lite emulation gives $\sigma = 0.027$ (C2, wave 2, 5e93769089). In AeST it can only be lowered, and the same BBN line gives $K_B < 0.0778$ (C2).
- μ_K **(v3's dust mass): no allowed value.** Growth needs at least 411 h/Mpc (M1), BBN with the literal K needs 1.84×10^8 h/Mpc (Q6), and galaxies cap it at 14.8 h/Mpc; no allowed value fixes clusters (S4).
- u **(the retained fraction):** a measured, convention-dependent function of scale with no mechanism (PAPER58, D6, E1, E4).

9 Tests

9.1 Galaxy tiles (P4)

The static limit is the law itself (AQUAL, $\mu = x/(1+x)$; C4, PAPER58 E2). P4 recomputed five galaxy tiles with v3's static limit on baryons only (external field solved, screening at 24 and 90 pc; no settling, supply, catchment, edge or round rules).

- **SPARC RAR: CONDITIONAL.** One Υ per galaxy, rms 0.054 / 0.054 dex; median Υ 0.52 / 0.44 (the alternative falls below the frozen 0.50–0.80 window). The χ^2 line fails on both footings (63–64% of galaxies acceptable); the record's $\nu_{\rm mono}$ does the same.
- **MeerKAT local a_0: PASS** (+0.033 dex against SPARC's same recipe). **Groups: PASS if isolated** (+0.010 / −0.010 dex), ABOVE LAW at $g_e = 0.03\,a_0$ (Z 4.7); no group-field estimate is on disk.
- **Milky Way vertical force, as recorded statically: FAIL.** AQUAL's extra field is disc-like: $K_{z,1.1} = 92.3$ / 96.9 against 66.4 ± 2.2 (χ^2 178 / 251 against a line of 67.5); v3 has no round rule. Under v3's own dust dynamics (U1) it is STRUCTURAL in the literal theory and history-dependent for multi-stream dust.
- **Tidal dwarfs and DF2/DF4, as recorded statically: FAIL.** With the host field solved and the phantom kept they sit −0.59 / −0.62 dex and −1.66 / −1.74 (ln) below v3. U1 reverses this: they are dust-free and pass natively (Section 9.5).

9.2 The Milky Way and the seals (S3)

- **Outer slope: SHAPE EXCLUDED, robust.** The law's slope at 15–27.5 kpc is −0.126 / −0.114 against the data's −0.325; $Z = -4.37$ / −4.62, worse than the record's round rule (−3.63 / −3.90). Every ingredient tried fails, screening is inert, and the μ_K^2 term would make the slope shallower.
- **Circular speed (U1).** $v_c(R_0) \leq 216$ / 221 km/s in every branch, against 229 ± 4.6.
- **Seals.** v3's Gaia DR4 Milky-Way file `SEALED_V3_MW_DR4.json` (sha256 `15b7e6ff`…; the earlier CFG553 file untouched), with a new observable P6: the external field's in-plane asymmetry of V_c, 3.0 / 5.2 / 8.2% at 20 / 30 / 60 kpc. v3's ALMA AQUAL-disc reading `SEALED_V3_ALMA_AQUAL.json` (sha256 `b5d65a5e`…).

9.3 Cassini and the planets (C6, D7d)

Criteria 1b2ced55a4, result ae428dc252.

- **The literal form passes Cassini.** All 24 cells ($L_\star$ = 24, 50, 90 pc; both footings; four Galactic fields) pass Hees et al. (2014) and Park et al. (2026). At 24 pc $Q_2 = 0.93$–$1.86 \times 10^{-34}\,{\rm s}^{-2}$; at 90 pc 1.76–3.52×10^{-36}. The margin below the Hees line is 4.8×10^7 to 5.1×10^9.

- **Mechanism.** Inside $L_\star$ the Sun's field is a constant hedgehog $GM/L_\star^2\,\hat{r}$ with no quadrupole. Removing the curl stiffness (D7d used it) lowers Q_2 by 9–19%; it does not open a quadrupole.
- **The frozen rule reads UNDECIDED.** One validation check (C2) fails at 3.8×10^{-11} against a 10^{-12} line: rounding in $(\kappa^2/2 + 1) - 1$ at $\kappa = 0.01$. The identity itself is exact in sympy. The script exits 1 (7/8).
- **Planets.** The anomaly is the hedgehog $GM/L_\star^2$: at 24 pc Mars is at 0.173 of its bound, Saturn 0.035 and Earth 0.028.
- **The unscreened law, for contrast,** fails Cassini by 2.44–5.19× under AQUAL (PAPER58, E2).

9.4 Strong field (P3)

- **PASS on all five, two scoped.** Black holes, static and non-rotating only: Schwarzschild with the khronon on Painlevé–Gullstrand leaves and $\mathbf{P} = 0$ solves every v3 equation; the static foliation that the weak-field law uses is singular at $r = 2M$, so regular holes need a tilted, infalling khronon. Shadow, ISCO and photon-sphere shifts are at most 2.9×10^{-11}; ringdown shifts at most 2.8×10^{-11}; neutron stars $dR/R \leq 2.7 \times 10^{-28}$.
- **Strong coupling: PASS as an estimate.** The worst loop factor is 1.3×10^{-3} at Eöt-Wash (38.6 μm, $L_\star = 90$ pc). The lab strong-coupling scale is 3.5–7 μm, a margin of only 1.7 in k, so sub-5 μm gravity tests would probe it.
- **The propagating pair.** Where $D \cdot P > 0$ the khronon/dust pair propagates: about $10^{-4}\,c$ in the Solar System and $0.44\,c$ at a neutron-star surface. P3 left $D \cdot P < 0$ open; H4 closed it as ILL-POSED.

9.5 Hosted small systems and v3's own dust (S1, U1, T1, E3, G2, E6, P1)

- **With v3's external field solved (S1).** The dwarf satellites sit +1.33 / +1.30 dex above v3 at the $L_\star$ floor (+1.72 / +1.68 at 90 pc). This supersedes the satellites' $+0.89 \pm 0.13$ dex above the screened prediction (PAPER58, E3), which used isolated screening without the host field; the external field alone costs +0.58 dex. Coma's UDGs go from undecided to FAIL: +1.12 / +1.08 dex at $z_{\rm sys}$ +4.3 to +4.7. Both rest on equilibrium in the present host field.
- **What v3's own dust can supply (U1).** Theorem T1: held dust never exceeds the AQUAL phantom (14,373 equilibria; largest ratio 0.666). Theorem T3: held dust is weightless in a static host, so a system that falls with its host carries no dust and is Newtonian. Consequences under v3's dynamics: satellites +2.00 / +1.99 dex, STRUCTURAL; Coma's UDGs +1.41 dex ($z_{\rm sys}$ 5.4), STRUCTURAL; tidal dwarfs ($+0.055 \pm 0.114$ dex) and DF2/DF4 a NATIVE PASS, with no knob; cluster cores STRUCTURAL at Blanchet and Skordis's own μ_K. Field galaxies keep the phantom at the cap (relative speed at most 1.8 km/s). Group members depend on whether group dust is held, and the Ursa Major members contradict the dust-free branch (12.6–14.4σ per galaxy). What would have to change: the dust coupling (and J).
- **One external-field strength does not fit (T1): FOOTING SPLIT, FRAGILE.** With one multiplier ϵ on v3's solved field, the canonical footing prefers a weak external field ($\hat{\epsilon} = 0.02$, 95% interval [0, 0.26]); the alternative is not decided ($\Delta\chi^2$ 7.96 against a line of 9); goodness of fit $p = 0.023$ / 0.014. Coma's UDGs want $\epsilon \leq 0.21$ / 0.24 and P2's lensing $\epsilon \geq 0.67$ / 0.81. A weaker external field cannot rescue the satellites: at $\epsilon = 0$ they still sit +0.91 / +0.89 dex above, and their residual rises with the host field at every ϵ (slope $+0.76 \pm 0.26$ at $\epsilon = 0$).
- **A pre-infall retention rule does not close it (G2, 92716d0148 / 3cdcf3b4c5):** on native inputs the offset stays $+0.92 \pm 0.14$ dex (canonical) / $+0.89 \pm 0.15$ (alternative), because the retained cold energy pays the law's phantom first. It closes only with a ΛCDM subhalo mass put in by hand, and then overshoots 8 systems.
- **The framework's dynamics cannot fix a satellite's catchment** (E6, c42925d9dc / 0bea588820): it is set by the patch's initial overdensity and infall epoch. Median $z_{\rm req} = 11.28$; the required catchment is an ordinary 1.21σ fluctuation in an externally calibrated initial field.
- **Supply from the satellites' chemistry** (E6, P1). E6's leaky box (their [Fe/H], no ΛCDM input) bounds the catchment from below at a median 38.33 times the present baryons, so the median need is 0.211 of the supply; 6 of 36 tested systems are short at central values. P1 replaced E6's premise ($\zeta = 1$ winds, no pre-enrichment) with the satellites' metallicity distributions: the premise fails (10 systems too narrow), but the supply survives at the median (ratio of need to supply 0.463 / 0.445); it is not robust at 2σ in the widths (median 1.12 / 1.08 with every width at its 2σ-low end); pre-enrichment raises the supply. Supply is not mechanism: nothing binds the cold energy inside the half-light radius (G2, U1).

9.6 Cluster cores (F1, H2, S4): a real failure for both forms

F1 (9fd5aee16e / f1803b0256) rebuilt the eRASS1 × KiDS cluster-core stack from the raw data with its own KiDS reader and estimator (326 clusters; it reproduces the record's stack within 1σ in every bin).

- **The law on the clusters' baryons fails.** In the primary rebuild (AQUAL, native h, gas rescaled, miscentring) $A_{\rm MID} = 1.68$–1.72 (canonical, $Z = +5.6$ to $+6.2$) and 1.54–1.58 (alternative, $Z = +4.9$ to $+5.5$); with the boost sign fixed (post-result) 1.81–1.85 / 1.66–1.70. The field equation, native h, gas calibration, R_{500} and miscentring each move it by at most 4%. At the corner most favourable to the law it is still 1.33–1.37 (canonical) and 1.22–1.26 (alternative), and that corner pushes the outskirts below the law.
- **No added cold profile fixes it:** any profile that fixes MID overshoots the outskirts. The theory's dust inside 1 Mpc is not derived.
- **AeST's static dust cannot supply it (H2, 364a56c250 / 385f61a14c):** its mass term ramps across 0.3–1 Mpc and oscillates negative outside, where the data need a flat excess then none. Its best profile χ^2 is 27.7–39.6, against 27.3–41.8 for the law alone and a pass line of 11.07.
- **No allowed μ_K fixes them in v3 (S4): SCOPED NO-GO.** In spherical symmetry the Khronon static equation is H2's AeST system with $\mu \to \mu_K$ (reproduced to 0.0058). No allowed (μ_K, zero point) passes even one cluster cell (0 of 1,804); the best profile χ^2 is the law's own, 33.0 / 27.5 against 11.07. Small μ_K ($\leq 0.1\,{\rm Mpc}^{-1}$) leaves the clusters unchanged to 1–2%; $\mu_K^{-1} = 1$ / 0.3 / 0.1 Mpc gives $A_{\rm MID} = 1.33$ / -3.84 / -9.86 and wrecks galaxies first: outer lensing $\Delta\chi^2$ +110 already at $0.1\,{\rm Mpc}^{-1}$, because the oscillation sets in at a radius $\propto M^{1/6}$, the reverse of the needed ordering. The absolute dust is negative over 46–89% of the cluster volume for $\mu_K \geq 0.3\,{\rm Mpc}^{-1}$.
- **Three postulated additions also fail:** a potential-depth selector, which neither action contains (J1); a braneworld Weyl term, for every closure (K1); and that term's tidal-charge closure screened by $L_\star$, which never reaches cluster scales (M2).

9.7 Galaxy lensing (F2, P2, S2)

- **Outer lensing, photometric isolation (F2, e9816b47d2 / 7c8c3d5302).** The frozen rule reads REAL FAILURE, but the frozen sample leaks satellites' hosts, so the reading is UNDECIDABLE. The record's $\Delta\chi^2 = +100.1$ came from a catchment supply edge the covariant theory lacks; without it the far-outer amplitude falls from 2.45 to 0.96 of the data (canonical). With the weakest on-disk field the amplitude (data over model) is 4.45. The isolated stack's outer signal moves by 0.74 ± 0.09 (8.6σ) with the isolation cut.
- **Outer lensing, spectroscopic isolation (P2): UNDECIDABLE (N too small).** 2,251 GAMA host-vetoed lenses separate zero-field and catalogue-field v3 by only 0.87σ / 0.96σ; about 22,000–26,500 are needed. The data sit 2.8–3.0σ below zero-field v3 and are consistent with the catalogue's own field ($y_e \approx 6.6 \times 10^{-5}$), so F2's "tolerates only $y_e \lesssim 3 \times 10^{-5}$" does not survive spectroscopic isolation; T1 traces it to F2's host-leaky sample.
- **Inner (blue) and red lenses (S2): both CONDITIONAL.** Blue: data/v3 0.88 (canonical, PASS but fragile) / 0.80 (alternative, 2.9σ short); the hot circumgalactic fraction and $M_\star$ decide it. Red: the inventory alone is 1.31 / 1.20 short; with CFG667 and the IMF α–σ posit, 1.02 / 0.93, consistent.

9.8 Outer globular clusters (D10, T1)

Each cluster was solved as a Plummer sphere in the Galactic field with the literal form's own equations. The unscreened law ($L_\star = 0$) is excluded in all 18 cells ($\Delta\chi^2 = 4.9$–22.8), and no finite $L_\star$ beats Newton. So the globulars give a lower bound only. Pal 4 and Pal 14 dominate; the bound is limited by the stellar M/L of Pal 4. They need screening, not the external field: at $\epsilon = 0$ the floor would rise from 21.7 to 28.4 pc (T1).

9.9 Binary pulsars, gravitational waves and ALMA (C8, P3, D4, S3)

Binary pulsars and GW170817: no constraint; tensor speed exactly c (Section 4). ALMA: expected unscorable on both dates (gate S/N 0.04–0.58 against a required 5). The plans are sealed; addendum E (ffb14e40bc) adds the $n = 1$ and $n = 2$ kernels as labelled parallel readings beside the $\nu_{\rm mono}$ verdict kernel, and the harness has the kernel switch (92fa9f9471). S3's v3 AQUAL-disc reading needs +0.03 to +0.05 dex more $V_{\rm bar}^2$ than the algebraic kernel at GS4; addendum F (owner-approved, pre-data, f6f0ba66a4) adds it as a labelled parallel reading, and the verdict kernel stays $\nu_{\rm mono}$.

10 Sealed predictions for data still to come

- **Gaia DR4 wide binaries (20 kAU).** Screened: $\gamma_v - 1 = 8.1 \times 10^{-6}$ / 1.9×10^{-6} / 5.8×10^{-7} at $L_\star = 24$ / 50 / 90 pc (C6). Unscreened law-first (AQUAL): 1.167 (canonical) / 1.199 (alternative) at the primary external field. Scored on the whole-catalogue $\hat{\gamma}$ under the owner's pre-data $\sigma_{\rm sys}$ rule (pre-DR4 note amendment A3, 3d82af0a7e): definition (b) primary, (a), (e), (g2) must agree, else UNDECIDED; footings scored separately. Expected power at 30,000 pairs under (b), from DR3 (non-scoring): 3.21–3.96σ canonical.
- **Gaia DR4 Milky Way:** v3's sealed file and P6 asymmetry (S3, above).
- **ALMA U4** (public 2026-11-06) and **GS4** (2026-11-28): frozen CFG806/CFG809 plans plus addenda E and F; expected unscorable.

11 What v1.0 left pending, and what is pending now

- **Native CMB: no longer pending.** v3: Q6 and Q7 (Section 6). v4: C7's milestone 5 (Section 7), on the code that passed its validation (7/7, milestone 4, b0ced915c8) and reproduces pristine CLASS in the exact dust limit to 2.2×10^{-6}. None of these tests the galactic law: a_0 is absent from the background and from linear order (C2: the invariant that carries a_0 vanishes on FRW at first order), so the CMB cannot test the galactic law.
- **Native cosmic shear (CFG960): no longer pending, and not about v3.** The canonical 1024^3 run finished on the engine of amendment A6 (f72bef68ea) and was scored as frozen: KiDS EXCLUDED, $\Delta\chi^2 = +47.28$. It tests the record's retired settling engine, not v3; the alternative run was not run (the chain was stopped to free memory for L1). The record's canonical settling rules were already excluded by raw KiDS (CFG596, $\Delta\chi^2 = +68.7$).
- **Pending: v3's nonlinear growth (L1).** L1 has built and validated a Khronon-native particle-mesh engine. Its production runs are in progress; no result is given here. Because literal v3 has no radiation era (Q6), any L1 result belongs to v3 on a dust background with a yet-unspecified early K.

12 Conclusions, and what this is not

- **The covariant candidate is ill-posed at saddles.** v3 is ill-posed wherever the khronon density is negative: every two-body saddle and the external-field lobes around every system (H4). The γ_L term that screens the Solar System drives it. v4 is ill-posed wherever the aether twists (I1) and runs away on FRW for the dust shape tried (C7). Neither form is a complete, healthy theory.
- **v3 fails several structural tests:** linear growth at the allowed μ_K (F3, M1, Q6, Q7); a radiation era in its literal form (Q6); hosted satellites and Coma's UDGs (U1); the Milky Way's outer slope (S3); cluster cores for every allowed μ_K (F1, S4). The Milky Way's vertical force and circular speed are in tension (P4, U1).
- **What passes:** Cassini, the planets and the outer globulars (screened); the strong field, scoped; the MeerKAT a_0; groups if isolated; tidal dwarfs and DF2/DF4, natively. SPARC is conditional; κ is consistent but not discriminating; galaxy lensing is conditional or undecidable.
- **What the data require of any successor:** a cold component that binds in hosted systems and in clusters, and grows like dust. v3's held dust does neither: it is weightless in a host (T3), never exceeds the phantom (T1), cannot supply cluster cores at any allowed μ_K (S4), and does not grow at the cap (M1, Q6, Q7).
- Not a UV completion: an effective theory with a preferred foliation.
- Not derived constants: κ, $L_\star$ and the dust abundance are fitted; the free function J_x and the stiffness term are postulated.
- The CMB cannot test the galactic law at linear order.
- Criterion A (no influence faster than light) is violated, by a real signal too small for any test identified; for v4, I1 adds a criterion-B violation in principle.
- The dwarf supply is conditional (chemistry assumptions, no binding mechanism); under v3's own dust the satellites are a structural failure.
- No particle species is added: the extra mass is the cold-energy sector, and the dust here is a field sector. No result here is evidence for the framework over its rivals.

Disclosures. C3 scores 9/11 with two frozen fails kept and a post-result amendment (6dba29d773). C5's three frozen fails (H5, H6, H9) are kept; its first complete run (6/10) had two code bugs, fixed with no pass line changed. C6's frozen verdict is UNDECIDED (float cancellation); its earlier executions and a 1.04 GB mutation run are

disclosed. C8's first run (6/8) failed on rounding and a solver limit, and peaked at 2.3 GB; fixes changed no pass line. D10's amendment A1 is pre-result. E6's amendment A1 is pre-result and A2 post-result. The Khronon equations come from the arXiv HTML via the C4 lane and are not verified on disk; the AeST sources are on disk (C2 provenance). Rebuild lanes: F2's frozen REAL FAILURE is kept beside its post-result UNDECIDABLE reading; G2 scores 4/6 (two MC-comparison fails kept); H2's frozen one-sided rule was corrected post-result; H3's frozen claim fails on three lines and its K5 fail is kept; F3's amendment A1 is post-result. Lanes added in v1.1: Q6's frozen V2, V3 and V5 validation gates failed, and its native numbers rest on post-result gates (amendment A3); Q6 and Q7 read Blanchet and Skordis via the arXiv HTML (not on disk; provisional). P3 scores 17/18 (frozen B1 kept as FAIL). H4 scores 13/14 (frozen convergence line C2 kept). U1 scores 15/16 (N4 fails as frozen on grid resolution). S3 scores 28/31 (three mis-specified frozen checks kept). S4's frozen rule reads UNDECIDED (its K5 and K7 lines fail, neither about the physics); its CANNOT is under amendment A1, written after test runs and before production. C7's amendment A4 is post-result. L1's frozen KB check failed and is re-scored under a post-result statistic (A2). The pre-DR4 note's amendment A3 repeats a statement about the alternative footing that PAPER58 v1.1 withdrew; the scoring rule does not depend on it.

Changes in v1.1 (2026-10-23). v1.0 is doi:10.5281/zenodo.23585319. (1) Erratum M1: v1.0 attributed v3's linear growth failure to $J'(0) = -1$ making the Newtonian term vanish on FRW (abstract, growth section, v4 section); the cause is Blanchet and Skordis's cap on μ_K, and J_x holds the dust better than $J'(0) = 0$. F3's gap numbers stand. (2) New results from the v3 recompute lanes, with their verdicts as written in the theory draft (commit be52681b9b): Q6, Q7, C7 (milestone 5), H4, I1, P3, P4, S1, U1, T1, S2, P2, S3, S4, P1, J1, K1, M2; L1 is in progress with no result. (3) Superseded: the satellites' +0.89 dex as the screened theory's gap (now S1 and U1); "v4 is the better-surviving candidate" (I1, C7); "only tensor modes propagate" and the unqualified WELL-POSED (P3, H4); F2's tolerated field (P2, T1); E6's chemistry premise (P1). (4) The v1.0 pending items are resolved: native spectra exist (Q6, Q7, C7), and CFG960 was scored on the retired settling engine. (5) Title and abstract rewritten to state these results plainly. All other v1.0 results are unchanged.

Repository: `covariant_push_2026_10_22/` (C1–C8, `THEORY_V3_DRAFT.md`); `rebuild_2026_10_22/` (F1–U1, L1); `doors_push_2026_10_22/` (D10, E6; D2, D4, D6, D7d, E1–E4 via PAPER58); `prep_2026/gaia_dr4_prep/` (pre-DR4 note, A1–A3). Companion notes: PAPER58 v1.1 (doi:10.5281/zenodo.23579379), PAPER57 (doi:10.5281/zenodo.23532885), PAPER49 v1.2 (doi:10.5281/zenodo.23533323). Theory: L. Blanchet, C. Skordis, JCAP 11 (2024) 040 (arXiv:2404.06584); L. Blanchet, C. Skordis, arXiv:2507.00912 (2025); C. Skordis, T. Złośnik, PRL 127, 161302 (2021) (arXiv:2007.00082); M. Milgrom, arXiv:1605.07458. Bounds: A. Hees, W. M. Folkner, R. A. Jacobson, R. S. Park, PRD 89, 102002 (2014); R. S. Park et al., PRD 114, 024066 (2026). AI assistance: Anthropic's Claude (lanes, review, drafting).